

# HBSM: Hybrid Boundary Sensitivity Mechanisms

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## Abstract

This is a living paper shell for a preregistered Mac-first experiment. HBSM tests whether sensitivity in frontier hybrid reasoners can be explained and predicted cheaply. The broad forward-only sensitivity claim is wounded by recent related work, so no novelty or positive result is claimed yet.

## 1 Current Gate

The branch must pass sensitivity heterogeneity, no-forward predictor, and mechanism gates. If it overlaps with HORN, it should be folded into a unified hybrid-quantization paper.

## 2 Mac Artifact Status

A deterministic synthetic B1/B2 packet validates the layer-sensitivity and cheap predictor readout. It reports kurtosis-versus-sensitivity Spearman rho 0.657 and two boundary-flagged top-decile hits, producing `SYNTHETIC_PASS_REAL_LAYER_SENSITIVITY_NEXT`. This is synthetic-only: it is not model evidence, not a novelty claim, and not GPU evidence. The metadata-only model eligibility packet identifies the current live hybrid targets and architecture hashes, but no corresponding weights are cached repo-locally. HBSM therefore has no real forward-sensitivity evidence on this Mac yet.

## 3 Next Evidence

The next admissible row is a real layer-sensitivity sweep on current hybrid reasoners. If cheap predictors fail to correlate with forward sensitivity, HBSM should be killed or folded into HORN rather than kept as a separate paper.

## 4 Admissible Real Packet

The real B1 packet must use fixed boundary flags from the shared architecture maps. Each row must report `model_id`, `layer`, `boundary_flag`, `precision_perturbation`, `kl_or_nll_drift`, `cheap_predictor`, `parameter_count`, `weight_norm`, and `control_type`. The packet must include `architecture_map_hash`, `summary.md`, `matching_row_count`, and pass `experimental.shared.check_gate_packet --mode real --project hbsm`.

| Required baseline        | Purpose                                         |
|--------------------------|-------------------------------------------------|
| Random top-decile flags  | Tests enrichment beyond chance.                 |
| Layer-index baseline     | Controls depth effects.                         |
| Parameter-count baseline | Controls layer size.                            |
| Weight-norm baseline     | Controls simple magnitude predictors.           |
| Boundary-only baseline   | Tests whether HORN already explains the effect. |
| Train/test split         | Reduces cheap-predictor overfit.                |

Table 1: Controls required before HBSM can claim a cheaper sensitivity predictor.